Ti Beta-C Spring Development Testing for Low Cycle, Cold Deep Space Applications

Pavlina Karafillis * and Grady Lynch*

Abstract

Titanium springs are an attractive option compared to stainless steel springs due to their superior energy density. There is a lack of data for Titanium springs compared to their stainless counterparts, particularly at cold temperatures. To address potential performance risks of Titanium springs at cold temperatures on the Mars Sample Return Program, testing was performed. Mechanical cycling and life testing up to 3000 cycles were conducted at ambient temperatures and at -135°C. No strength loss, yielding, or rupture was observed at ambient or cold test temperatures. A stiffness increase of approximately 5.7% was observed between ambient and cold testing. Destructive microstructure evaluation of test and control springs demonstrated that alpha-enriched grain boundaries were present before and after testing. The alpha-enriched grain boundaries did not appear to function as crack initiation sites or cause any crack propagation as a result of the mechanical cycling. The processes utilized in Titanium helical compression spring fabrication were assessed to understand potential sources of alpha case and alpha-enriched grain boundary formation. From this assessment, a methodology for eliminating alpha-enriched grain boundaries is proposed.

Introduction

Stainless steel helical compression springs, particularly 300 series and 17-7 alloys, are commonly used in space mechanism applications. While not as common, Titanium springs are an attractive option due to their improved energy density, allowing for lower mass and improved packaging efficiency with equivalent performance. Ti 13V-11Cr-3Al springs were used in various aerospace applications throughout the 20th century but encountered significant challenges with production and fabrication. Ti 3Al-8V-6Cr-4Mo-4Zr, also known as Ti Beta-C, was developed in the 1960's as an easier-to-produce alternative to Ti 13V-11Cr-3Al. Since then, Ti Beta-C has been used in many aerospace applications, though little documentation exists for its usage at cold temperatures. A thorough literature survey was conducted but yielded no documented uses, test data, or material properties for Ti Beta-C at temperatures below -70°C. On the Sample Retrieval Lander Aeroshell, part of the Mars Sample Return mission, the protoflight cold temperature for the springs will be -135°C. To address this temperature gap, a development test campaign was carried out to exonerate three risks tied to Ti Beta-C helical compression spring usage at -135°C. First, the concern of reduced strength at cold was addressed by mechanical cycling at cold and ambient. Second, the concern of reduced performance due to fatigue at -135°C was addressed by a life testing up of 3000 cycles (over 100x expected total cycles) at -135°C. Lastly, the concern of crack initiation and growth at protoflight cold was addressed by microstructure analysis.

The testing described herein was designed to evaluate the performance of Ti Beta-C helical compression springs at -135°C and demonstrated that Ti Beta-C helical compression springs are acceptable for use in low cycle, cold deep space applications. Additionally, an investigation into alpha case formation and the associated crack initiation and growth risks was conducted, yielding a proposed method for effectively mitigating said risks.

Test Set Up

Test Overview

Three springs formed from Ti Beta-C per AMS4957, henceforth individually referred to as the Unit Under Test (UUT) and shown in Figure 1, were cycled. Each UUT is a nominally 254-mm (10-inch) long Ti Beta-

* Lockheed Martin Space, Littleton CO

C spring with a 10.64-mm (.419-inch) nominal wire diameter and a 78.64-mm (3.108-inch) nominal outer diameter. The UUT were not the same size and dimensions of the proposed flight springs but were brought to stress states matching the flight springs in order to conduct a representative test.

Figure 1. Titanium Spring

Each UUT was exposed to the test flow described in Figure 2. One control spring from the same lot as all the UUTs was sectioned to evaluate its microstructure.

Figure 2. Test Flow

<u>Load Frame Testing</u>
Load frame testing was conducted at ambient and cold temperatures. As shown in Figure 3, the UUT was installed in custom 17-4PH fixturing and a CRES threaded rod. During all load frame testing, the UUT was loaded in compression, while the load frame maintained tension. All testing was completed on a 97860-N (22-kip) Load Cell and Load Frame from MTS Systems, calibrated to ±1% accuracy. Cold test temperatures were achieved using a Watlow Thermal Chamber and Temperature Controller and type E thermocouples, which can be seen in Figure 4. In the load frame, data was collected at about 100 Hz.

During load frame ambient characterization, the three UUTs were individually mechanically cycled five times at 22°C and 41% relative humidity. Mechanical cycling was executed between the free length of 248.9 mm (9.8 inches) and working height of 248.9 mm (6.2 inches) in a sine-wave profile with a maximum crosshead speed of approximately 127 mm/sec. Note that since an initial of compressive force of 90-130 N was maintained throughout the test, the spring was always slightly compressed, meaning it did not truly return to its free length on each cycle.

Following ambient characterization, each of the three UUTs were individually mechanically cycled five times at the cold temperature of -135°C to -145°C. The same crosshead speed as ambient testing was used. The Watlow temperature controller was set to a ramp rate of -5 degrees per minute to reach cold, and +5 degrees per minute to return to room temperature after the test. The test article dwelled at the set point temperature for 5 minutes prior to cycling.

Following cold characterization, each of the three UUTs were individually mechanically cycled 3000 times at the cold temperature of -135°C to -145°C. The previously noted crosshead speed and thermal chamber ramp rates and dwell times were used. For life testing, thermocouple 2 as shown in Figure 4 was omitted.

Figure 3. Load Frame Fixturing

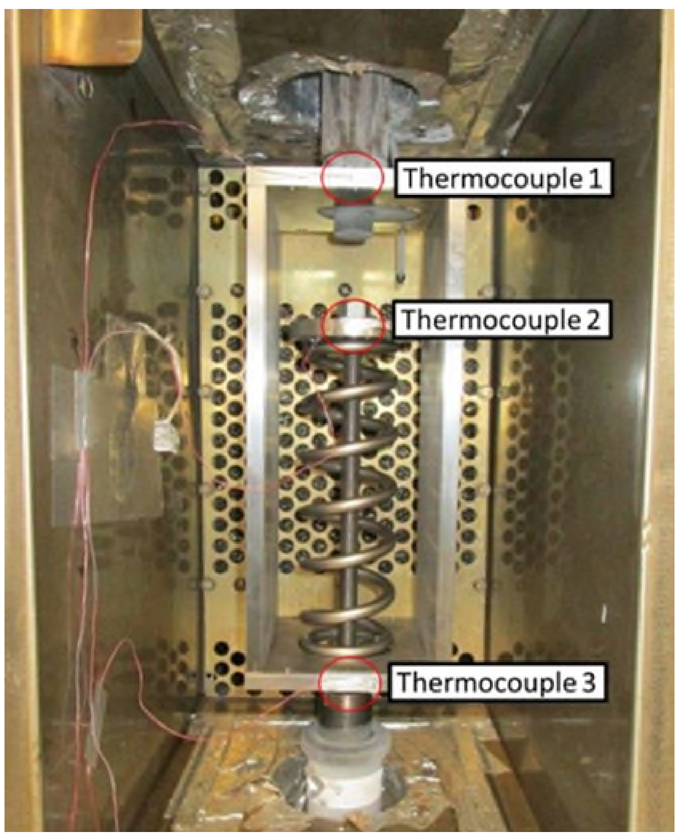

Figure 4. Thermocouple Placement

Microstructure Assessment

Following load frame testing, the three UUTs plus one control spring were each sectioned in three locations from the active coils. Microstructure evaluation was performed on each of the 12 sections using standard micro-sectioning techniques. This includes SiC abrasive paper, diamond polishing compounds, and AIO slurry. Then, the samples were etched to bring out the microstructure. Finally, the sections were imaged using a Zeiss Axio Observer Metallograph.

Test Results

Achieved Coil Stress

Table 1 summarizes the achieved max compression forces for each UUT during each test, as well as the corresponding maximum coil stress, using as nominal dimensions of the UUT. The maximum force achieved correlates to the maximum coil stress using Equation 1, where *P* is the maximum spring force, *D* is the mean spring diameter, *d* is the wire diameter, and K_s is the shear stress-correction factor. Assuming material properties per AMS4957 and a shear stress calculation per distortion Energy Theory, the allowable ultimate shear stress on this spring is at least 716.4 MPa (103.9 ksi).

$$\tau_{max} = K_s \frac{8*P*D}{\pi*d^3} \tag{1}$$

Table 1. Achieved Coil Stress Summary

Serial Number	Test	Max Force (N)	Max Coil Stress (MPa)	Ultimate Shear Stress (MPa)
SN02	Ambient Characterization	3126.3	495.3	716.4
SN03		3082.8	492.0	
SN04		3137.5	500.7	
SN02	Cold Characterization	3229.6	511.6	
SN03		3303.5	527.2	
SN04		3213.2	512.8	
SN02	Cold Life	3343.3	529.6	
SN03		3313.3	528.8	
SN04		3312.0	528.6	

Cold vs Ambient Performance Changes

As expected, the data across all tests are linear, with all datasets having best fit lines with R^2 values greater than 0.999. The consistency of the stiffness throughout the >3000 cycles demonstrates that no performance degradation was observed due to fatigue. As a detailed spot check, the spring stiffness derived from only first 100 cycles and the last 100 cycles of the life test were calculated to be within 87 mN/mm (<0.3%) of each other in all cases and R^2 remained greater than .999 in all cases. The y intercepts shown on the best fit lines reflect that the springs were preloaded to between 90 and 130 N and the initial position was zeroed out at this load prior to cycling. Not all the y intercepts of the best fit lines match the load at the zeroed position. This variation is attributed to the small but typical non-linearities in the first 5 mm of stroke of the spring, and measurement error.

Table 2 compares best fit spring stiffness values across ambient characterization, cold characterization, and cold life tests. There is a clear stiffness increase at cold temperatures compared to ambient, which agrees with the inverse correlation of temperature and Young's Modulus presented in the Aerospace Structural Metals Handbook entry for Ti Beta-C, although, as previously mentioned, no explicit data exists for the temperature range of interest. The average magnitude of stiffness increase from ambient to cold was 5.7% and the cold spring stiffness was 1.4 N/mm (8 lbf/in) greater than ambient with 95% confidence. Equation 2 demonstrates that Young's Modulus is directly proportional to spring stiffness. Accordingly, the

5.7% measured stiffness increase can be attributed to a corresponding 5.7% Young's Modulus increase under the cold test temperatures. Taking 104,110 MPa (15,100 ksi) as the value for Ti Beta-C Young's Modulus at ambient conditions, as is represented in the Aerospace Structural Metals Handbook, the calculated Young's Modulus at cold test temperatures is 110,047 MPa (15,961 ksi). In Equations 2 & 3, *k* is the spring constant, v is Poisson's Ratio, D is the mean diameter, N_{ac} is the number of active coils, *d* is the wire diameter, and *E* is Young's Modulus.

$$k = \frac{d^4 * G}{8 * D^3 * N_{ac}}, G = \frac{E}{2(1+\nu)} \tag{2}$$

Solving for E:

$$E = k * \frac{16 * D^3 * N_{ac} * (1+\nu)}{d^4} \tag{3}$$

For all three springs tested, the spring stiffness observed during cold life testing was slightly higher than the spring stiffness observed during cold characterization testing. This was not expected because the measured temperature was within ±5°C between runs, and the life testing was not consistently colder than the characterization testing. For SN03 and SN04, the change between the two cold tests is negligible, and well within the ±1% calibration accuracy of the load cell. The spring stiffness change between cold tests for SN02 is notably higher than the other two springs, but still within the ±1% load cell calibration accuracy. Finally, the p-value for a two-tailed T-Test, between the stiffness of cold characterization testing and life testing is 0.49, indicating that there is no statistically significant difference between the two means. As the apparent stiffness changes between cold characterization and cold life testing are not statistically significant, and the application being investigated was not sensitive to variations less than ±1%, root cause for slightly increased stiffness seen in test at extended cold was not identified.

Table 2. Ambient vs Cold Spring Stiffness

Serial Number	Ambient Char. Spring Stiffness	Cold Char. Spring Stiffness	Cold Life Spring Stiffness	Percent Change Between Ambient and Cold Life	Percentage Change Between Cold Ambient and Cold Life
	[N/mm]	[N/mm]	[N/mm]	%	%
SN02	32.204	33.720	34.167	6.1	1.3
SN03	32.046	33.873	33.910	5.8	0.1
SN04	31.802	33.361	33.457	5.2	0.3
Average	32.0	33.7	33.8	5.7	0.6
STD DEV	0.20	0.26	0.36	0.46	0.66
Max Delta	0.4	0.5	0.7	0.9	1.2
Variance	0.0	0.1	0.1	0.2	0.4

Spring Performance Summary

Figures 5 through 7 plot the force vs displacement data of all three UUTs at cold and ambient, with best fit lines. No strength loss, yielding, or rupture was observed during mechanical cycling at ambient or cold. This mechanical cycling addressed the concerns of reduced strength at cold, and reduced performance due to fatigue.

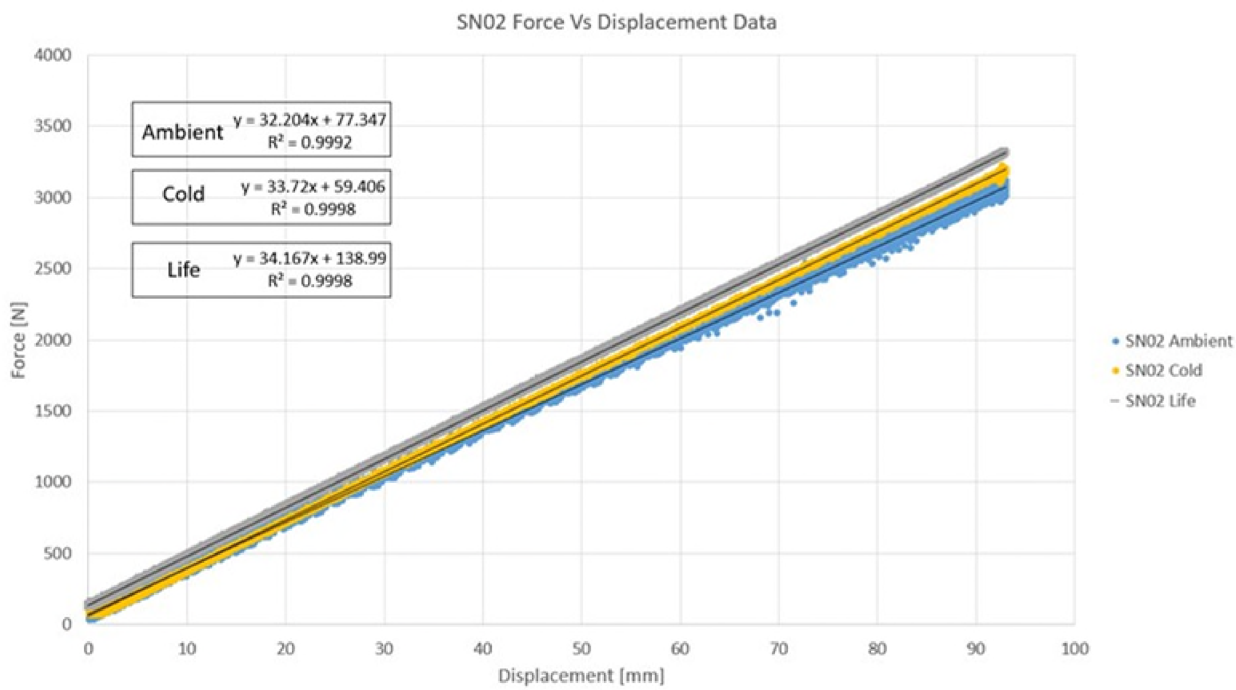

Figure 5. SN02 Force vs Displacement Overlay

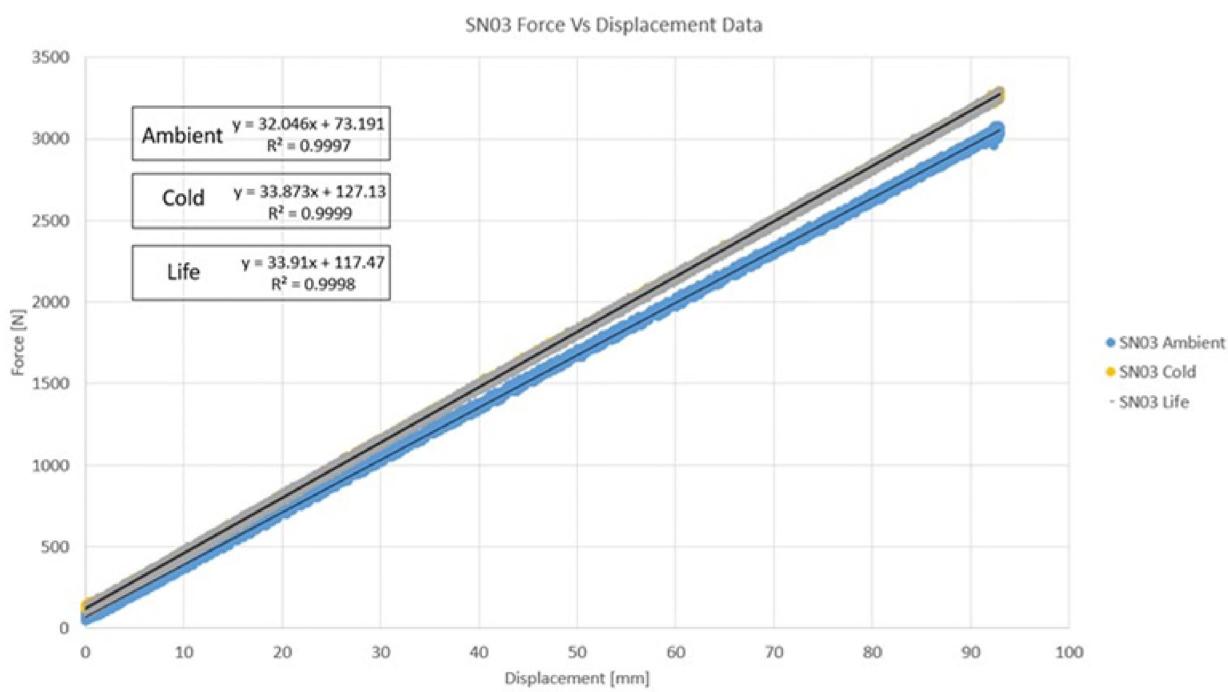

Figure 6. SN03 Force vs Displacement Overlay

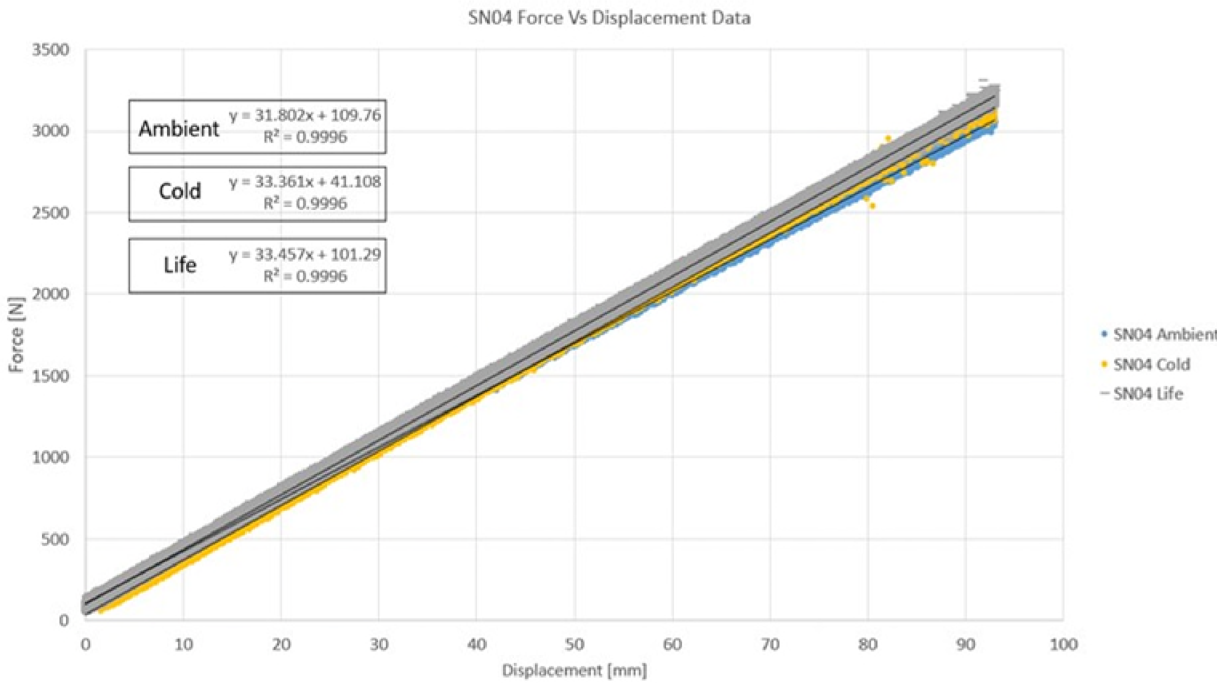

Figure 7. SN04 Force vs Displacement Overlay

<u>Microstructure Evaluation</u>

The overall microstructure was found to be consistent across all sampled springs. The primary finding from microstructure evaluation was the presence of alpha-enriched grain boundaries and small cracks on the outer edges all spring sections evaluated. The observed cracks are too small to be resolved during a level 3 dye penetrant inspection and are confirmed to have formed prior to the mechanical cycling testing, since the control spring exhibited the same cracking. See Figures 8 and 9 for observed alpha-enriched grain boundaries and micro cracks in a UUT and a control spring. The fact that the cracks on the mechanically cycled springs did not appear larger than those on the control spring suggests that the mechanical cycling at cold temperature did not cause crack growth. Moreover, the performance of the springs during characterization and life cycling suggests that even if flight springs had similar microcracking, the springs would still function as designed.

While the test results indicate that the springs would function as intended with alpha-enriched grain boundaries, their unexpected observation and known risks of cracking associated with alpha case prompted a deeper investigation into the formation mechanism for alpha case and the available options for removal of the alpha case and alpha-enriched grain boundaries.

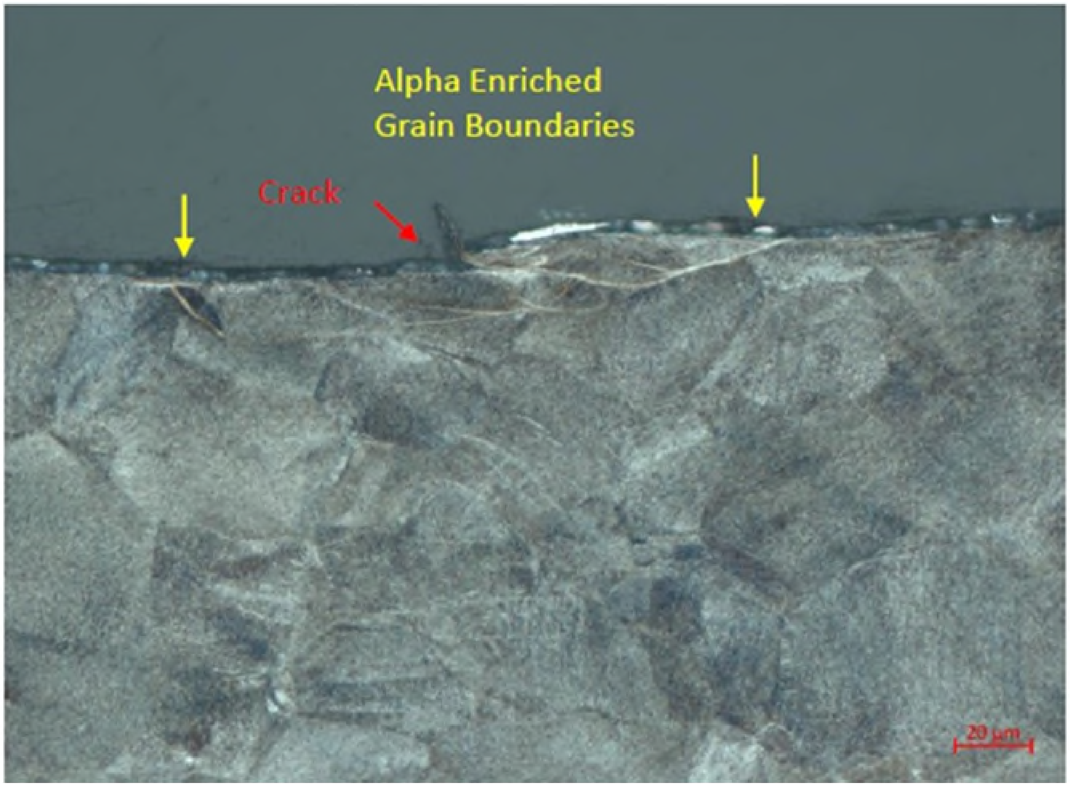

Figure 8. SN 02 Microstructure Evaluation (UUT)

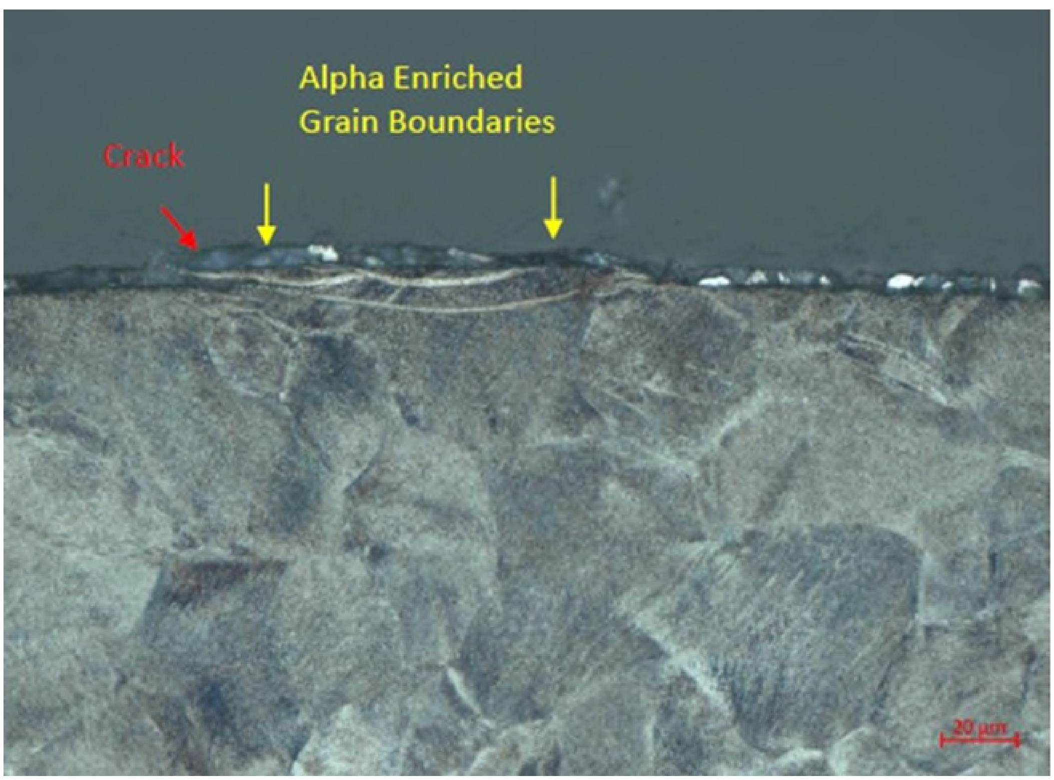

Figure 9. SN 05 Microstructure Evaluation (Control Spring)

Alpha Case Development and Crack Mitigation

After identifying alpha-enriched grain boundaries during UUT microstructure evaluation, an assessment of alpha case formation and mitigations during Titanium helical compression spring manufacture was conducted. Figure 10 summarizes the manufacturing process and highlights steps of interest where alpha-enriched grain boundaries form and are subsequently removed. Note that the term “alpha case” will be used to henceforth describe both alpha case and alpha-enriched grain boundaries. These two physical phenomena are similar and related, but not identical. Alpha case is the more generic term, while alpha-enriched grain boundaries specifically indicates that grain boundaries are enriched with alpha case.

As can be seen in Figure 10, alpha case is understood to form both during the initial raw material solution treat step and during the subsequent coiled spring age heat treat step. Alpha case formation and subsequent removal resulting from the initial raw material solution treat step is governed by AMS4957 and carried out by the raw material vendor. The subsequent age heat treat and alpha case removal step, which is controlled by the spring vendor or a subcontractor thereof is critical to the final state of alpha case on the part.

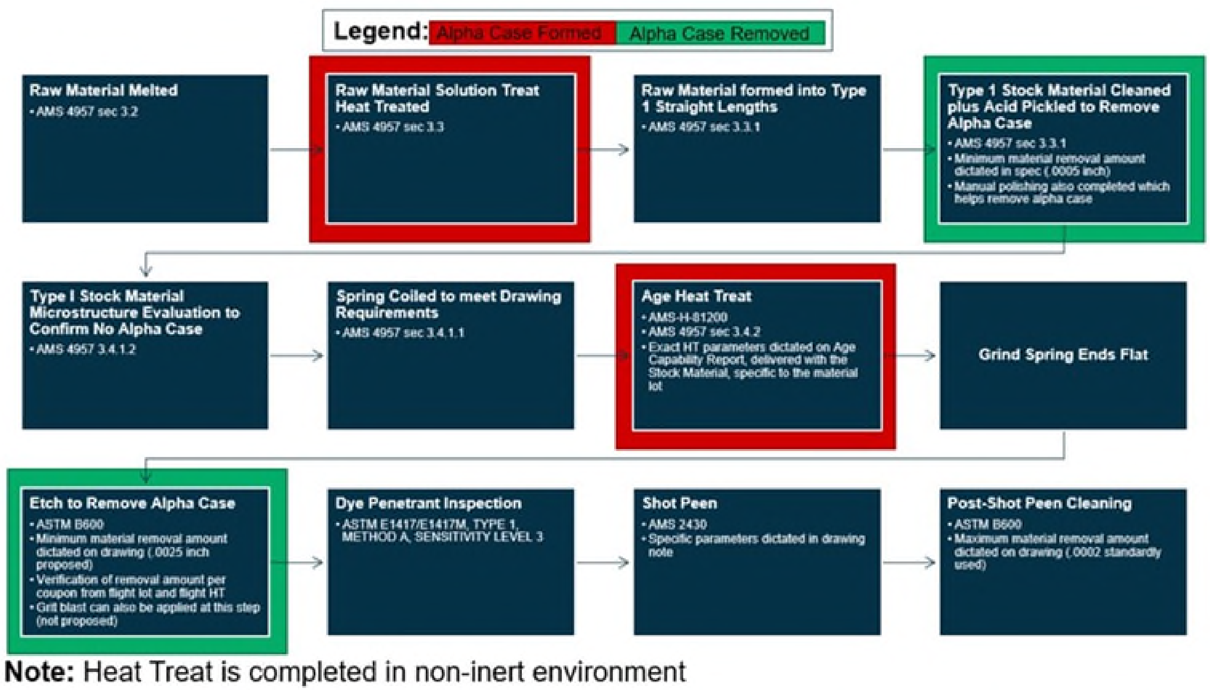

Figure 10. Ti Beta-C Spring Processing Steps

<u>Alpha Case Formation</u>

In order to understand the phenomenon of alpha case and how it forms on a Titanium part, it is instructive to review the Technical Datasheet for Ti-3AL-8V-6Cr-4Mo-4Zr published by Carpenter Technologies [1]. Carpenter Technologies explains that alpha case is a “brittle oxygen-stabilized alpha layer” of Titanium that forms on the exterior of the part during heat treat in a non-inert environment. While completing heat treat in an inert environment may decrease alpha case formation, it is typically financially advantageous to complete heat treat in a non-inert environment. Regardless of the heat treat environment, Carpenter advises that “it is very important to completely remove not only the surface scale, but the underlying layer of brittle alpha case as well” to preclude surface crack-induced failure. Moreover, “in [titanium] beta alloys, alpha case tends to penetrate the grain boundaries beneath the surface. Thus, the actual depth may be greater than is readily apparent, and the effective thickness may not be uniform.” Variable alpha case thickness was noted in the UUTs and control spring sampled.

Quantification of Alpha Case

Precise quantification of alpha case can be completed via sectioning and microstructure evaluation using a coupon of the same wire lot that undergoes the same heat treat process as fabricated springs, or with a spare sacrificial spring.

In leu of sectioning and microstructure evaluation, a minimum material removal amount can be implemented that bounds expected alpha case formation depth. It is understood that the amount of alpha case developed and corresponding minimum material to be removed varies based on heat treat parameters of temperature and time, which inherently varies between material lots. AMS4957 and dictates an acceptable range of temperature and time during which heat treatment may occur in order to achieve the required mechanical properties. Despite these variables, published data does exist for recommended material removal depth for Titanium Alloys based on heat treat temperature and time [2] [3]. Given the typical heat treat time of up to 10 hours at up to 593°C (1100F) and published data in [2] and [3], a removal value of 25.4 μm (.001 inch) is recommended for Titanium alloys. [3] All micro cracks identified in the control and UUTs were within 25.4 μm of the surface and would have been eliminated with an etching of 25.4 μm. However, the depth of the alpha case was seen to be as far as 48.3 μm in the springs sampled. No sources were found to explicitly call out Ti Beta-C in material removal tables, but [2] identifies that the alpha case depth may vary with alloy type, possibly explaining the deviation from published data seen. In order to bound the material removal completed in the manufacturing process of the sampled springs, a minimum material removal for Ti Beta-C springs of 63.5 μm (.0025 inch) is proposed in the post etch heat treat for Ti Beta-C springs for use on springs in this application. It is important to note that 63.5 μm is very small in comparison to wire diameter, as the UUT has a wire diameter of 10.64 mm (.419 inch), and the proposed application being assessed has a wire diameter of 12.7 mm (0.5 inch). Further work is recommended to assess if the etched amount should vary with the wire diameter.

Methods of Alpha Case Removal

The Carpenter Technologies tech data sheet [1] recommends that “Surface removal can be accomplished by mechanical methods such as grinding or machining, or by descaling (using molten salt or abrasive) followed by pickling in a nitric/hydrofluoric acid mixture”. At least two methods for alpha case removal currently employed in industry have been identified, abrasive grit blast and Hydrofluoric etch per ASTM B600 have been identified for alpha case removal. Vendor experience also indicates that shot peening after alpha case removal is a prudent step to further mitigate surface cracking risk. Further work is required to understand if one of these methods provides better results.

Verification of Alpha Case Removal

Similar to the question of determining how much alpha case is present, the most robust approach to verification of alpha case removal is via sectioning and microstructure evaluation, as is required following raw material solution treat heat treat per AMS4957. That said, it is often impractical to require coupon sectioning and microstructure evaluation in the middle of spring fabrication to verify complete alpha case removal. An acceptable alternative is to calculate material removal depth of 63.5 μm (.0025 inch) on a representative coupon via mass measurement pre- and post- alpha case removal operation. This approach works as-follows:

1. Prepare multiple representative coupons along with flight parts. Coupons must be from same material lot and must experience same age heat treat as flight parts.
2. Precisely measure mass of each coupon before alpha case removal operation (i.e. grit blast or etch)
3. Expose coupon to alpha case removal operation. Parameters for this operation must be identical to parameters used for flight parts.
4. Precisely measure mass of coupon after alpha case removal operation.
5. Use mass delta to calculate material removal depth over surface area of part

If desired, this mass measurement approach can be supplemented by final sectioning and microstructure evaluation of a sacrificial spring to develop additional confidence in complete alpha case removal. If alpha case is identified at this time and an additional etch is executed to remove it, the shot peen and later steps must be repeated for their impacts to take effect.

Conclusion

The successful test campaign conducted on Ti Beta-C helical compression springs demonstrates acceptable performance to temperatures as low as -135°C. The testing demonstrated a Ti Beta-C stiffness increase of 5.7% between ambient and cold test temperatures. The tested springs had residual alpha enriched grain boundaries and microcracking prior to testing yet still showed no signs of fatigue-induced failure or performance degradation, further supporting the conclusion that Ti Beta-C is appropriate for low cycle, cold space applications. While testing and microstructure analysis of UUT and control springs indicates that cycling up to 3000 cycles causes no identifiable increase in crack growth and crack initiation, the alpha case removal minimum depth of 63.5 μm (.0025 inch) and subsequent verification steps presented herein are believed to be sufficiently robust to effectively mitigate the risk of crack-induced failure for low cycle springs.

Acknowledgements

We would like to thank Julia Pimentel for all her work conducting testing on the load frame, and Curtis Ricotta for conducting the microstructure analysis. We would also like to thank Jeanne Eha for providing valuable material science input on the test formulation and when investigating the alpha enriched grain boundaries. Lastly, we would like to thank Mason Woish, Mike Knopp, and Don Lewis for their input into understanding and mitigating the alpha enriched grain boundaries. This testing was conducted by Lockheed Martin Space as part of contract number 1664478 with the Jet Propulsion Laboratory.

References

1. Carpenter Technologies Ti-3Al-8V-6Cr-4Mo-4Zr (Beta C™) Tech Data Sheet. https://www.carpentertechnology.com/hubfs/7407324/Material%20Saftey%20Data%20Sheets/Ti%203Al-8V-6Cr-4Mo-4Zr.pdf
2. Donachie, M. J*. Titanium: A Technical Guide – Second Edition.* ASM, ©2000.
3. AMS2801 Rev B